

EXPERIMENT 4

HILL CIPHER

AIM

To implement the Hill Cipher substitution technique using Python.

ALGORITHM

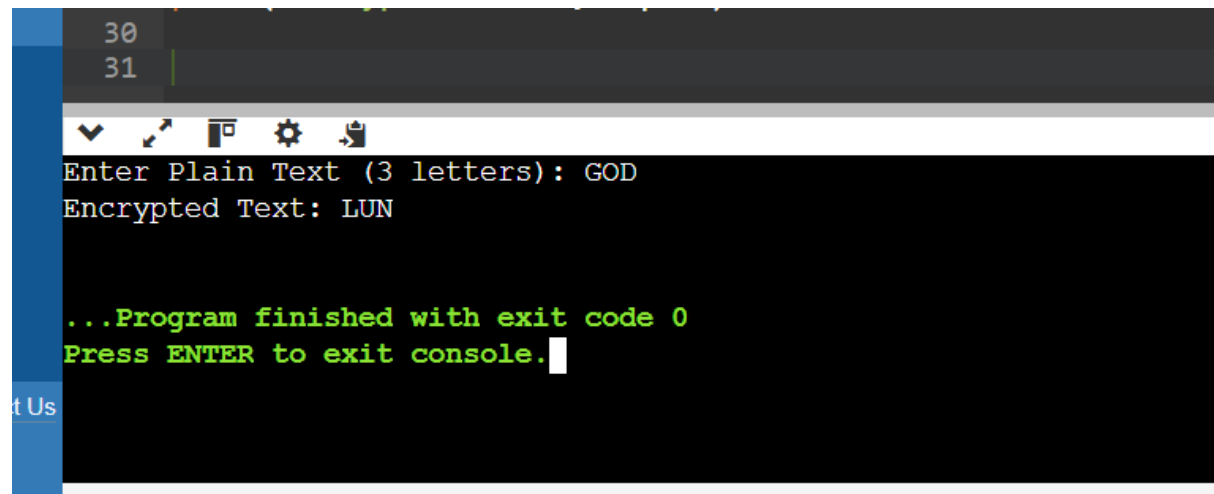
1. Read the plaintext from the user.
2. Read the 3×3 key matrix.
3. Convert the plaintext into numerical values (A=0, B=1, ..., Z=25).
4. Multiply the plaintext vector with the key matrix modulo 26.
5. Convert the result back to characters to obtain the cipher text.
6. Display the encrypted text.

PYTHON CODE

```
key = [  
    [6, 24, 1],  
    [13, 16, 10],  
    [20, 17, 15]  
]  
  
text = input("Enter Plain Text (3 letters): ").upper()  
  
while len(text) < 3:  
    text += "X"  
  
text = text[:3]  
  
P = [ord(ch) - 65 for ch in text]  
  
C = []
```

```
for i in range(3):  
    total = 0  
    for j in range(3):  
        total += key[i][j] * P[j]  
    C.append(total % 26)  
  
cipher = ""  
  
for num in C:  
    cipher += chr(num + 65)  
  
print("Encrypted Text:", cipher)
```

OUTPUT



```
30  
31  
Enter Plain Text (3 letters): GOD  
Encrypted Text: LUN  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

RESULT

Thus the Hill Cipher substitution technique was implemented successfully using Python.